

The Transport Layer II: TCP

Reliability, Connections, and Flow Control

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Where We Have Been (The Recap)

Before we start today's topic (TCP), let's look at the "family photo" of protocols we have met so far.

Week 2: HTTP (The Web / Text Messages)

Week 4: UDP (Streaming / "Fire & Forget")

Upgrade

Week

Week 4: ICMP (Ping / Diagnostics)

Today, we are moving back up to Layer 4 (Transport) to meet the "Responsible Big Brother" of UDP.

Recap 1: The HTTP Header (Text)

Protocol: HTTP (Layer 7)

Style: Human-Readable Text.

The "Header" Structure

It's just lines of text! There are no bits or bytes to count.

```
GET /index.html HTTP/1.1  
Host: www.google.com  
User-Agent: Chrome
```

Pros: Easy to debug.

Cons: Slow to parse (computers hate reading text).

Recap 2: The UDP Header (Tiny)

Protocol: UDP (Layer 4)

Style: Fixed Binary (8 Bytes).

The Structure (Only 4 Fields!)

Source Port & Dest Port

Length & Checksum

Why so small?

- It doesn't care if you get the packet.
- It doesn't track order.
- It just sends data. Fast.

Recap 3: The ICMP Header (Diagnostics)

Protocol: ICMP (Layer 3)

Style: Error Reporting.

The Structure

Type (8=Ping) & Code (0=Req)

Checksum & Unused

Purpose:

- Not for moving data.
- Only for saying "Hello?" (Ping) or "Error!" (TTL Exceeded).

Today's Hero: TCP

Definition

TCP (Transmission Control Protocol) is the standard for **Reliable** data transfer.

The "Big Brother" of UDP:

- **UDP:** "I sent it. Good luck."
- **TCP:** "I sent it. Did you get it? No? I'll send it again. Are they in order? No? I'll fix the order."

Where do we use TCP?

- Web Browsing (HTTP needs TCP).
- Email (SMTP).
- File Downloads (FTP).
- *Basically anything where missing a single byte would break the file.*

The TCP Header: A Monster Contract

To be reliable, TCP needs a LOT more information than UDP.

Size: 20 Bytes (Minimum) vs UDP's 8 Bytes.

Source Port	Dest Port
Sequence Number (Where does it fit?)	
Acknowledgment Number (Did I get it?)	
Flags (SYN, ACK, FIN)	Window Size

Two Critical New Fields:

- 1 **Sequence Number:** "This is byte #100 of the file."
- 2 **Acknowledgment (ACK):** "I received up to byte #100."

Concept 1: The 3-Way Handshake

Unlike UDP (which just shouts), TCP must "establish a connection" first. It's like a phone call.

Step 1 (SYN):

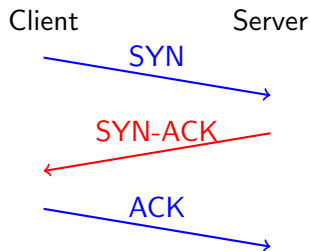
Client: "Hello? I want to talk."

Step 2 (SYN-ACK):

Server: "I hear you. I'm ready."

Step 3 (ACK):

Client: "Great. Here is my data..."



Concept 2: Reliability (The Receipt)

What happens if a packet gets lost?

The Rules of TCP:

- 1 Every time I send a packet, I start a **Timer**.
- 2 If the Server sends back an **ACK** (Receipt) before the timer runs out, great!
- 3 If the timer runs out (Timeout)... **I assume it was lost and send it again.**

This is "Flow Control"

TCP also stops you from sending too fast. If the server is overwhelmed, it tells you to slow down (using the "Window Size" field).

Summary: UDP vs TCP

UDP (Last Week)

- Header: 8 Bytes.
- No Handshake.
- Fast, Unreliable.
- Uses: Streaming, DNS.

TCP (This Week)

- Header: 20 Bytes.
- **3-Way Handshake** (SYN/ACK).
- Reliable (Retransmits lost data).
- Uses: Web, Email, Files.

Next Lab: We will use Wireshark to catch a TCP Handshake in action!